

Homework 4

due 08.11.2023, 14:00

Exercise 4.1 - Lattice Walk (10 points)

64 processes are arranged in an 8×8 lattice, with positions $\vec{x} = (x_0, x_1)$, $x_i = 0, 1, \dots, 7$, such that the rank of the process on lattice site $\vec{x}$ is given by $x_0 + 8x_1$. The lattice shall have periodic boundaries, i.e. $(x_0, 8) \equiv (x_0, 0)$ and $(8, x_1) \equiv (0, x_1)$.

The goal is to read a sequence of directions from the file `directions.dat` and, starting from rank 0, move through the lattice according to these directions and sum up the encountered process ranks along the way. The directions are encoded as integers 0,1,2,3 which stand for positive x_0 , positive x_1 , negative x_0 and negative x_1 direction. The final result is printed out by rank 0.

- Only rank 0 is allowed to do I/O operations.
- Every rank should know the ranks of its four nearest neighbors.
- No global (size 64) arrays are allowed.

The main loop may have this structure:

```
pos = 0
value = 0

loop:
  1) rank 0: read next line in file
  2) broadcast the direction, root: 0
  2) compute my neighbor's rank _n_ in this direction
  3) broadcast _n_, root: pos
  4) rank == pos:
      value += rank
      send value to rank _n_
  5) rank == _n_:
      receive value from rank pos
  6) pos = _n_
```